Binary to Excess-3 Converter

Theory

Binary to Excess-3 converter is a combinational logic circuit that converts a 4-bit Binary Coded Decimal (BCD) input into its corresponding Excess-3 code. The Excess-3 code is obtained by adding 3 (0011) to the BCD input. It is a non-weighted code and is self-complementing, which makes it useful in digital systems.

Task

Design and implement a Binary to Excess-3 converter using logic gates and verify the results with a truth table.

Mathematical Rule

Excess-3 Output = Binary Input + 3 (0011)

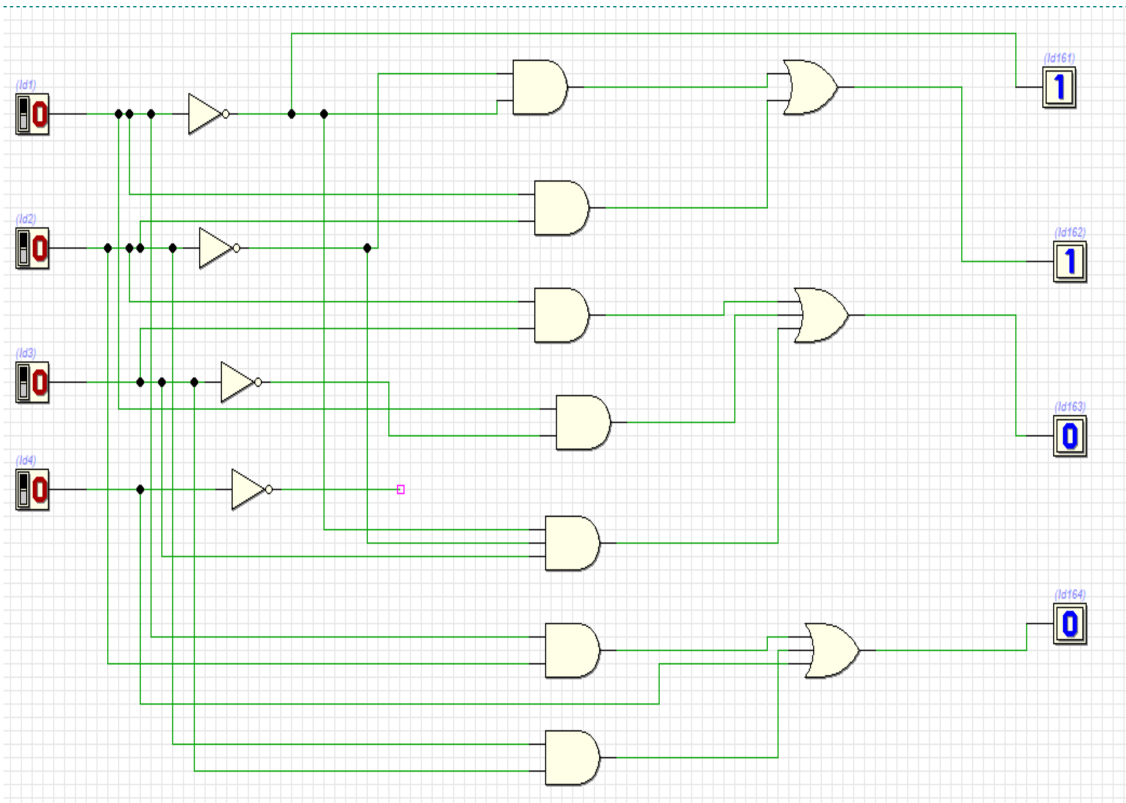
Truth Table:

Binary Input				**Access 3 Output**			
A	**B**	**C**	**D**	**W**	**X**	**Y**	**Z**
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

Working

The converter works by adding binary 0011 to the BCD input. For example, if the input is 0010 (decimal 2), adding 0011 results in 0101 (decimal 5 in Excess-3). The logic is implemented using combinational circuits derived from Karnaugh maps.

Diagram:



Procedure

1. Write the truth table for BCD to Excess-3 conversion.
2. Simplify the Boolean expressions using K-maps.
3. Implement the circuit using logic gates (AND, OR, NOT).
4. Connect the circuit on a breadboard using ICs.
5. Apply inputs and verify outputs.

Hardware Components

IC 7408 (AND Gates)
IC 7432 (OR Gates) IC
7404 (NOT Gates)
Breadboard, wires, power supply

Conclusion

The Binary to Excess-3 converter successfully converts BCD inputs into Excess-3 outputs. It is widely used due to its simplicity and self-complementing property.

Learning Outcomes

- Understand code conversion in digital systems
- Learn implementation using logic gates
- Gain knowledge of K-map simplification
- Understand practical circuit building